Sensor-Aided Doppler Estimation for Underwater Acoustic Links

Who estimates which Doppler, from what, at what cost — for group discussion

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12 October 2026 (results updated 13 October)

Target: IEEE JSAC special issue on Next-Generation Underwater Acoustic Communication Systems (deadline 31 March 2027)

CLAIM to be shown SIMULATED simulation result SUGGESTED design proposal OPEN needs material, a proof or an experiment SOURCE found by search, not yet checked in full

Main idea

The ground rule: estimation is the contribution

Contribution: the Doppler estimation — who estimates which part, from which information, how accurately, at what cost (pilots, bits, search).

Taken as a tool: the compensation. Once the estimate is right, resampling along the estimated trajectory works.

Supported by field data: on a surface buoy, a good frame-level estimate plus standard compensation leaves a residual of 0.2 subcarrier spacings (Lee et al. 2026). The open question is how to get the estimate right where that is not enough, and cheaply.

Doppler of a link

$$\epsilon(t) = \frac{\big(v_A(t) - v_B(t+\tau)\big) \cdot u}{c}$$

transmitter share + receiver share + slow share + path-specific share

changed on 12 Oct 2026; the compensation-focused deck is archived

Each part of the Doppler is known best in a different place

Part	Known best by	Reaches the receiver as
transmitter's motion during emission	transmitter's IMU, live	side information in the same packet (E1), or applied at the transmitter
receiver's motion during reception	receiver's IMU, live	direct input
slow part, sensor drift, direction errors	the received signal (preamble, postamble, pilots)	signal estimates
surface-scattered, path-specific	nobody	a floor

Today every UWA receiver estimates *all* of it from the received signal alone. **Proposal:** estimate each part where it is known and fuse them at the receiver (E2).

Phase 0: the transmitter's motion must travel with the packet

Real buoy motion	**2.3–2.4 s ahead**	**share $<$10 s waves**
CDIP Waverider (0.9 m), 3 Pacific sites	0.77–1.08	45–95%
OOI surface buoys (3.3 m, 4 t), Oregon, Irminger	0.76–0.93	70–94%

Normalised error of the best linear prediction of velocity (1 = no better than the mean). Every 0.8 s, 0.8 s ahead: 0.27–0.47.

SIMULATED Estimates one round trip old: -12 to -5 dB SIR at every range (0.5–4 km).

Consequences

- feedback of Doppler across a round trip fails (this killed our earlier forecast design)
- the transmitter's own motion, measured while it sends, is never stale if it travels *in the same packet*
- the receiver's own motion is measured live

The proposal: E1 side information, E2 fusion, E3 who estimates what SUGGESTED

E1 The transmitter carries its measured motion over the packet in the packet: a few quantised samples or coefficients. How many bits, and how the receiver reads them before it knows the Doppler.

E2 The receiver fuses E1, its own IMU and preamble / pilot estimates in one smoother over the packet. Sensors narrow the search and replace most pilots; the signal corrects sensor drift.

E3 Which part from which source, given sensor quality, latency, packet length and pilot budget (our placement results, reread).

Candidate claim SUGGESTED: the first sensor-aided Doppler estimation for UWA links — transmitter motion as in-packet side information, fused with the receiver's IMU and signal estimates; sized in bits and pilots; closed forms from estimation error to interference; real buoy motion.

Sending vs applying: the same transmitter estimate can be sent (E1) or applied (transmit-side pre-compensation); both are compared.

Gaps with the literature

Related work (1): Doppler estimated from the signal alone

Work	Estimation	Limit for us
Sharif et al., JOE 2000	Chirps at both ends of the packet	One scale per packet
Li et al., JOE 2008	Preamble scale + residual CFO	Frame-level; pilots for the residual
Mason et al., 2008	Multicarrier preamble: detection, sync, scale	Signal only
Cho–Kim, IET Commun. 2026	Constant acceleration from the cyclic prefix	$\approx$1 s frames
Sun 2022; Hu 2024	Kalman Doppler tracking (DSSS, FH-FSK)	Signal only, no motion prior
Lee et al. 2026 (preprint)	10.5 h surface-buoy trial: frame scale + per-symbol CFO	per-symbol tracking gains only $\times$1.15 in PER at $\approx$9 dB

All estimate everything from the received signal. Lee et al. show it is enough in a noise-limited link (QPSK, 0.9 s frames, sea state 3, one moving end): our regime must be stated against it.

Related work (2): sensors, side information, radio analogues

Work	Mechanism	What it lacks
Aubry 2019; Garin 2021; Choi et al. RA-L 2022	Modem Doppler into navigation filters	Runs the other way
DE-Sync 2018, APE-Sync 2019	Far end's Doppler returned in the reply	Only to set clocks
Oiwa et al., IEICE Tech. Rep. 2020	Accelerometer-aided compensation	One end; not peer reviewed
3GPP NTN; US 10,826,595	GNSS velocity $\rightarrow$ Doppler pre-compensation	Radio; ephemeris/GNSS
Gan, US 10,015,762 (UAV)	IMU $\rightarrow$ Doppler predistortion	Radio; patents
AsyMov, arXiv 2412.10387	Channel Doppler vs IMU (ISAC)	IMU helps only when paths are scarce

Local library, web and OpenAlex checked (9–12 Oct 2026): no UWA receiver uses motion sensors to estimate Doppler, and no transmitter sends its motion. Radio precedents are mostly patents.

Candidate gaps

G1 No UWA receiver uses motion sensors as an input to Doppler estimation

Sensor + Doppler work feeds navigation (Aubry, Garin, Choi); only Oiwa 2020 at one end. Radio: GNSS / IMU-aided pre-compensation (NTN, UAV patents).

G2 No transmitter sends its motion; side-information size not analysed

DE-Sync / APE-Sync exchange Doppler for clocks only; headers carry addresses, SNR, MCS.

G3 Accuracy / overhead / search trade-off with a sensor prior not analysed

Signal-only estimators (Li 2008, Mason 2008, Cho–Kim 2026, Kalman trackers); Lee 2026 maps the noise-limited regime only.

G4 (Extension) multi-user branches estimated one by one

Tu et al. 2011, 2013; per-user sensor priors unused.

Evidence so far (reused)

Sensor information is worth 4–15 dB at the same pilot budget SIMULATED

SIR [dB], 1 s packets	Seabed CDIP	Seabed OOI	Buoy–buoy
Signal only, one estimate per symbol	35.3	26.2	23.0
IMUs + one estimate per symbol	39.3	38.4	37.8
IMUs + quadratic over all symbols	**43.9**	**41.0**	**39.6**
IMUs + 2 estimates per packet (line)	39.8	37.1	36.0

Real buoy motion, 30 packets per point; IMU drift 0.01 m/s, 50 ms latency. With 20° path-direction error and 4 s packets, IMUs + per-symbol estimates keep 30–39 dB; 2 estimates per packet fall to 3–16 dB.

- same number of signal estimates, +4 / +12 / +15 dB with the sensors
- sensors cover the fast motion; the signal covers drift and what sensors miss
- this is E2 in its simplest form (sensor share + signal share, no proper fusion yet)

What the sensors must achieve SIMULATED

Factor	Sensors + 2 signal estimates	Sensors only
IMU drift 0.005 → 0.1 m/s	38–41 → 28–32	29–39 → 11
Latency 0 / 50 / 100 / 200 ms	42 / 37 / 30 / 20	32 / 33 / 28 / 16
50 ms, not bridged	25	17
Lever arm 2 m, corrected / not	38 / 13	31 / 3
Direction error 0 / 5 / 20°	37 / 30 / 18	30 / 13 / 0

SIR [dB]; buoy→seabed, OOI buoy.

Requirements on the sensor estimate

- drift: up to $\approx 10\times$ worse is fine when the signal corrects it
- latency $\lesssim 100$ ms (0.01 wave periods), bridged by the accelerometer
- lever arm from the gyro: mandatory
- path direction within $\approx 5^\circ$

Estimation error → interference, in closed form SIMULATED

Interference = phase variance of the residual within each FFT window:

$$I = (2\pi f_c)^2 \overline{\mathrm{var}_{\mathrm{win}}\Big(\int r\, dt\Big)}$$

for any linear estimator (weights from its estimates to the trajectory).

- a residual ramp costs 5× less than an offset of equal variance
- 85 simulated points: right winner at 81 (misses $\leq$0.6 dB); every strategy within $\pm$0.7 dB (median)

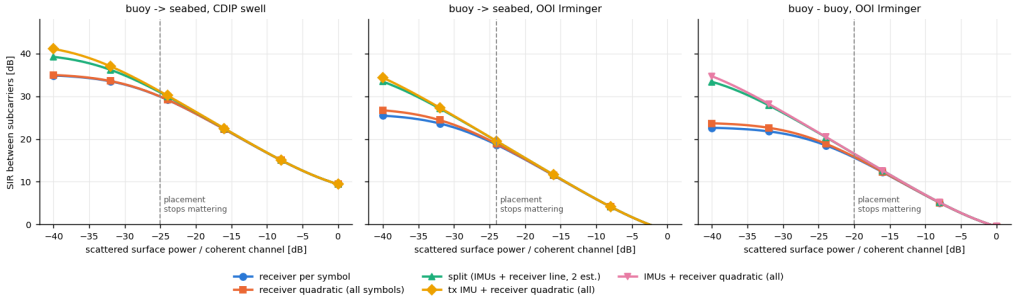
Role now: the metric that turns any estimation error (sensor, side information, pilots) into SIR.

Validated sources of error

- sensor drift, latency, lever arm, direction (node stage as a filter on the measured motion)
- the fit the receiver really makes from its estimates
- estimate noise through the fit's weights

Side-information quantisation (E1) and the fused covariance (E2) plug into the same formula.

A floor no estimate removes: surface-scattered Doppler SIMULATED



Better estimation pays only while the scattered surface power stays $\lesssim -20$ to -25 dB below the coherent channel (Bellhop with measured roughness, scattered power swept). OPEN literature value for real channels.

Proposed methodology

E1: Doppler side information SUGGESTED (built)

What: the transmitter's velocity along the path over the packet, from its IMU (lever arm, latency handled at the source), as K samples or polynomial / DCT coefficients, quantised to B bits.

Where: a robust low-rate header read after a coarse synchronisation (chicken and egg: the Doppler must be roughly known to read it), or carried by the preamble itself.

Output: residual Doppler and SIR vs B; header failure vs SNR; compared with applying the estimate at the transmitter (no bits, one intended receiver) and with the receiver alone.

Theory (T2): residual variance vs bits from the motion spectrum: representation error (polynomial / sampled kernels, done) + quantisation noise $\rightarrow$ interference (T1).

Expected size: a few coefficients per second of packet for wave motion (most energy below 0.3 Hz) — tens of bits, to be computed.

OPEN header design; IMU access at symbol rate on real modems.

E2: fusion at the receiver SUGGESTED (built)

Smoother over the packet (Kalman + RTS) on a wave-motion state: Doppler shares, rate, slow drift, IMU bias.
Inputs: E1 (with its quantisation error); own IMU samples (with latency); preamble / postamble / pilot measurements (SNR-dependent error).
Output: Doppler trajectory + covariance → resampler.

Gains to show: accuracy at high SNR and long frames; pilots and preamble saved for a target SIR; narrower Doppler search (acquisition at low SNR); robustness when the signal estimate fails.

Baselines

- Li et al. 2008 two-step (preamble scale + residual CFO), as in Lee et al. 2026
- per-symbol pilot tracking; signal-only Kalman
- Cho–Kim constant acceleration

Theory (T3): Bayesian Cramér–Rao bound with and without the sensor prior, vs SNR and pilots.

Results: the end-to-end chain

What we built and compared SIMULATED

Frame: up/down chirp — header (side information + CRC) — OFDM data (pilots, in-band nulls) — chirp. Rate-1/2 code, QPSK / 16-QAM, real pilot channel estimate, LLRs weighted by noise + interference on the nulls.

Channel: real buoy motion (CDIP swell, OOI Irminger), unknown start, noise; Bellhop multipath with each path's own Doppler from its angles.

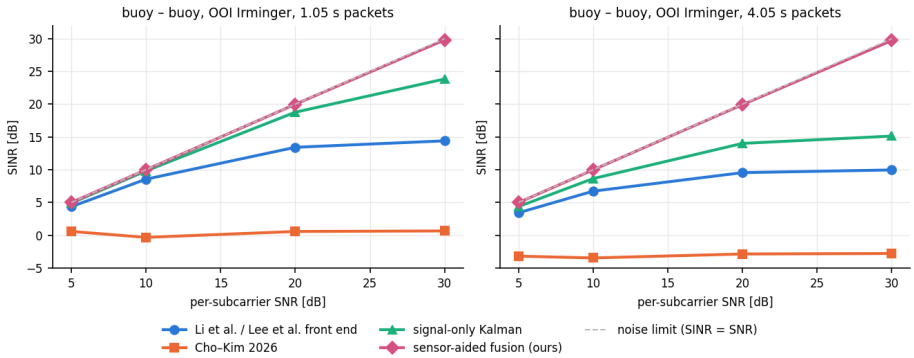
Ours: side information (16–96 bits) + own IMU → prior; Kalman fusion with postamble timing and per-symbol offsets; per-path cancellation with sensor-predicted Doppler.

Baselines (state of the art)

- Li et al. 2008 front end (as in Lee et al. 2026): preamble/postamble scale + per-symbol CFO
- Cho–Kim 2026: constant acceleration from the cyclic prefix
- signal-only Kalman: our smoother without sensors (the strongest signal-only receiver)
- Tu et al.-like multi-branch; signal-only cancellation (multipath)

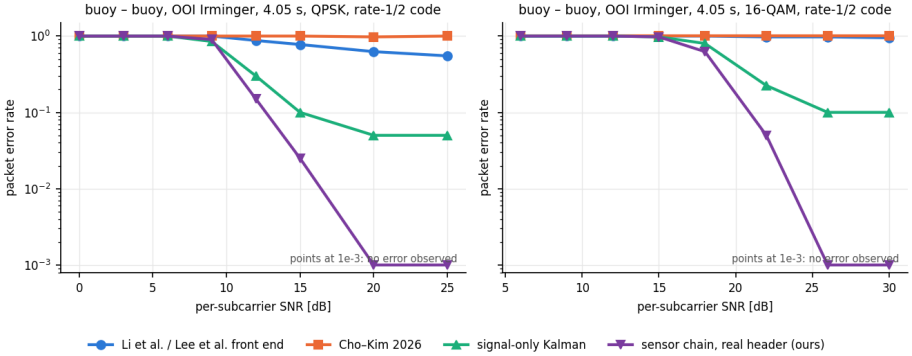
Two fixes made the comparison fair: chirp range–Doppler bias (up/down sweeps), and a wider CFO search for the baseline.

Single path: where the sensors matter SIMULATED



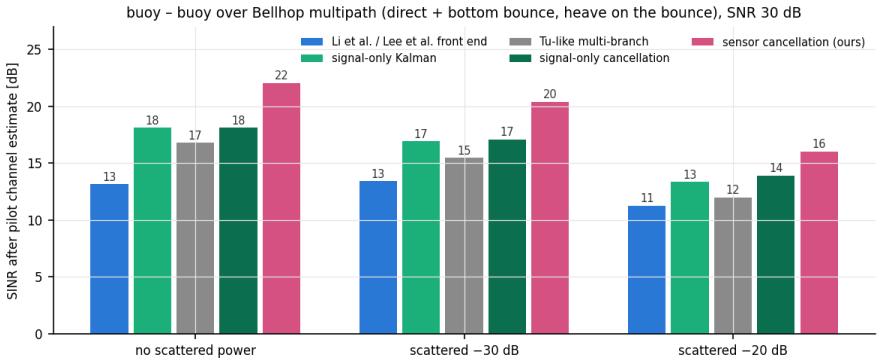
Gain of sensor fusion over the best signal-only receiver at 30 dB: CDIP 0 dB; seabed OOI +0.4 / +3.9 dB (1 / 4 s); buoy–buoy OOI +6.0 / +14.6 dB. Over Li/Lee up to +20 dB. 20 packets per point.

Coded PER: an error floor only signal-only receivers keep SIMULATED



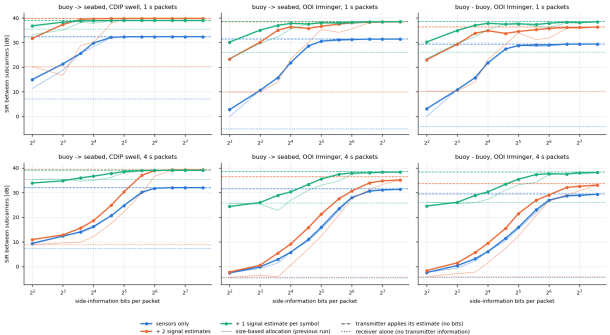
With 1 s packets, signal-only Kalman and our chain are equal (no errors from 15 dB QPSK / 22 dB 16-QAM); Li/Lee floors in the rough sea (4–6% QPSK, 23–55% 16-QAM). 40 packets per point; higher statistics running.

Multipath: per-path Doppler from sensors makes cancellation work SIMULATED



The bottom bounce carries both buoys' heave $\times \sin 10^\circ$: one resampling cannot fix two paths. Sensor-predicted cancellation: +3–4 dB over every signal-only receiver, PER 0.05–0.10 vs 0.20–0.30 at 20–30 dB. Needs path angles within $\approx$0.5–1°; delays from the signal.

Side information: a few tens of bits, read before the Doppler is known SIMULATED



- 10–30 bits per second of packet reach the gain of the transmitter compensating itself ($<$1% of the packet; pilots in every symbol: 25%)
- what to send depends on what the receiver estimates itself (stage-aware allocation, up to 2.7$\times$ fewer bits)
- header (DBPSK, CRC-8) read in 100% of packets at $\geq$5 dB, 95–100% at 0 dB (16–32 bits); then the chain equals ideal side information

Where we stand: claim and requirements

Claim SIMULATED: sensor-aided Doppler estimation removes the error floor that the best signal-only receiver keeps when the sea is rough, both ends move and packets are long, and makes per-path cancellation possible in multipath; it matches the best signal-only receiver elsewhere (Lee et al.'s regime).

Not yet: statistics for figures (running), multi-user, DESERT, writing.

Requirements

- IMU drift $\lesssim$0.01–0.1 m/s; latency $\lesssim$100 ms (bridged); lever arm from the gyro
- side information 32 bits (1 s) – 96 bits (4 s); +32 bits vertical for multipath
- path angles within $\approx$0.5–1°
- scattered surface power $\lesssim -20$ dB (else a floor for all)

Where sensor-aided estimation matters: our regime vs Lee et al. 2026

	Lee et al. 2026 (field)	**Where a better estimate pays**
SNR / modulation	$\approx$9 dB, QPSK + turbo	high SNR, 16-QAM and above (residual $0.2\,\Delta f \approx 9$ dB SIR is binding)
Frame length	0.875 s	$>$1 s (one model per frame cannot follow waves)
Sea / platform	sea state 3, small buoy, receiver only moving	rough seas, large buoys, both ends moving, AUVs
Overhead	preamble + pilots	short preamble, few pilots, narrow search
Acquisition	raw Doppler up to $0.6\,\Delta f$, near their failure point	sensor prior centres the search

CLAIM sensor-aided estimation helps where signal estimation is weak or costly; it must not lose elsewhere (AsyMov objection). Coded PER confirms it on Lee's own metric (Results).

Theory and evaluation

Theory

T1 estimation error → interference SIMULATED done

T2 residual vs side-information bits SIMULATED simulated; closed form open

T3 fused error and bound with / without sensor prior OPEN

T4 staleness: why feedback fails SIMULATED done

Evaluation

- real buoy motion (CDIP, OOI), modelled AUVs
- OFDM simulator, Bellhop channels, scattered-surface floor
- SIR, then coded PER; overhead (pilots, bits); search complexity
- network level: DESERT estimation-aware PHY

Risks

Noise-limited links (Lee et al. 2026): the frame-level estimate is enough there; state the regime, show coded PER.

Signal can beat sensors (AsyMov): fusion must gain where the signal is weak or costly and not lose elsewhere.

Chicken and egg: solved in simulation — header read in 95–100% of packets from 0 dB, fallback to signal-only otherwise.

Scattered-surface floor: above ≈ -20 dB, accuracy gains vanish.

Radio prior art (GNSS / IMU-aided Doppler, patents): claim "first for UWA, with real wave motion, side-information sizing and fusion".

Data: motion real, channels simulated.

Questions for the group

Questions for the group

- Which nodes have an IMU / INS, and can the modem read it at symbol rate with known latency?
- How many header bits per packet can side information take?
- Is transmit-side pre-compensation (applying the estimate) in scope next to side information (sending it)?
- Decided: SIR first, coded PER next — now urgent to answer Lee et al. 2026.